Task 1:

The first part of the project consists in designing the material and dimensions used in the making of a co-axial cable which will satisfy the customer's requirement.

Given informations :

- Device impedance = 75Ω
- Dielectric parameters ($\sigma = 0$, $\epsilon = \in r\, \epsilon o$, $\mu = \mu o$)
- Inner conductor radius = 0.5mm = 0.0005m

The ratio of the cable inductance per meter (L) and capacitance per meter(C) is given by $75 = \sqrt{\frac{L}{C}}$

It is thus necessary to find the formula for the inductance and the capacitance.

Step 1:

To find the inductance, it is useful to use Ampere law in order to find the magnetic flux density.

Calculations:

$$\oint \vec{B}\, d\vec{l} = \mu I_{en}$$

For an amperian loop of radius r, $a<r<b$; counting only inner current, $I_{en} = I$

$$B(r)(2\pi r) = \mu I \rightarrow B(r) = \frac{\mu I}{2\pi r}$$

The formula for the work done inside an inductor is $W = \frac{1}{2}LI^2$

By finding the energy dispensed inside an inductor, we could isolate the value of the inductor.

Calculations:

Energy per unit length:

$$W = \int u \, dV = \int_a^b \frac{B^2}{2\mu}(2\pi r)dr = \int_a^b \frac{\mu I^2}{8\pi^2 r^2} \cdot 2\pi r \, dr = \frac{\mu I^2}{4\pi}\int_a^b \frac{1}{r} dr = \frac{\mu I^2}{4\pi} \ln\left(\frac{b}{a}\right)$$

we know that: $W = \frac{1}{2} L I^2$

$$\frac{1}{2} L I^2 = \frac{\mu I^2}{4\pi} \ln\left(\frac{b}{a}\right) \longrightarrow L = \frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right)\left(\frac{H}{M}\right)$$

$$L = \frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right)$$

Now, we found that the value of the inductor is $L = \frac{\mu}{2\pi} ln(\frac{b}{a})$.

Step 2:

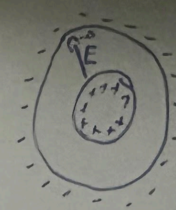
To find the capacitance formula, we assume that the inner conductor carries positive charges and the outer carries negative charges. From this assumption we can find the electric field that goes from the inner conductor to the outer conductor. We will then integrate the electric field to get the voltage between the inner and the outer conductor and substitute it inside the capacitor formula: $C = \frac{Q}{V}$

Calculations:

Finding Capacitance (C):

Assuming inner conductor carries +Q charges and outer carries Q^-

Finding $\vec{E}$ between the conductors:



$$\oint \vec{E}\, d\vec{A} = \frac{Q_{en}}{\varepsilon_0} \rightarrow \vec{E}(r)(2\pi r) = \frac{Q_{en}}{\varepsilon_0} \rightarrow E(r) = \frac{Q_{en}}{2\pi\varepsilon_0 r}$$

$$C = \frac{Q}{V}, \quad V = -\int_a^b E(r)\,dr = -\int_a^b \frac{Q_{en}}{2\pi\varepsilon_0 r}\,dr = -\frac{Q_{en}}{2\pi\varepsilon}\ln\left(\frac{b}{a}\right), \quad |V| = \frac{Q_{en}}{2\pi\varepsilon}\ln\left(\frac{b}{a}\right)$$

$$C = \frac{Q}{V} = \frac{Q}{\frac{Q}{2\pi\varepsilon}\ln(\frac{b}{a})} = \frac{2\pi\varepsilon}{\ln\left(\frac{b}{a}\right)}$$

Now, we found that the value of the capacitor is $C = \frac{2\pi\varepsilon}{ln(\frac{b}{a})}$.

Step 3:

Now that the formula for capacitance and inductance are found, we can substitute them in the initial equation $75 = \sqrt{\frac{L}{C}}$

$75 = \sqrt{\frac{L}{C}} = \sqrt{\frac{\frac{\mu}{2\pi}ln(\frac{b}{a})}{\frac{2\pi\varepsilon}{ln(\frac{b}{a})}}} = \frac{60}{\sqrt{\epsilon r}}ln(2000b)$ (we got 2000b from substituting the value of a with 0.0005m)

Step 4: Finding dielectric and conductor material and outer radius
For the dielectric material we want to choose a material that has the best efficiency/affordability ratio. For this we have two options.

Polyethylene: The dielectric constant for this material is 2.25 (F/M) and is an affordable option [1] . If using polyethylene, the outer radius will be approximately 3.33mm.
Polyvinyl chloride: The dielectric constant for this material is 3.18 (F/M) and is slightly more expensive than polyethylene[1] . If using polyvinyl chloride, the outer radius will be approximately 4.65mm.

For the conductor, we also want to use a material that has the best efficiency/affordability ratio. For this we have to choose between two options.

Copper: This material is as efficient as more expensive conductors such as silver while being relatively an inexpensive material[2].
Aluminium : This material is not as conductive as copper but is the cheapest option available[2].

We now have two interesting compositions for designing our cable, one favors efficiency and the other one favors affordability:

Polyethylene/Copper : By using polyethylene as a dielectric material, the outer radius of the cable will be smaller. Thus we will use a smaller amount of conductive material. This gives us the opportunity to use a more conductive material even if it is more expensive. Copper is the best option to be used here.
Polyvinyl chloride/Aluminium : Using polyvinyl chloride will result in a larger outer radius, so for this we will need to use a less expensive conductor to be more affordable. This is why using aluminium is the best choice.

Report:

Before doing this task of the project, I had a hard time understanding how to apply the concepts seen in class into a real life problem. I learned that doing calculations is only one part of electrical engineering, there is more to it like finding the best type of material to use in a particular context.

To solve the above problem, I had to search how co-axial cables work and why we need to apply inductance and capacitance to them. For this I went through various documentations ranging from Hydro-Quebec documents to youtube videos explaining how co-axial cables work.

References:

[1] "Dielectric Constants," HyperPhysics, Georgia State University. [Online]. Available: http://hyperphysics.phy-astr.gsu.edu/hbase/Tables/diel.html

[2]“Which Metal Is Best Conductor of Electricity? Top 5 Metals,” *Saryu Industries*, Aug. 08, 2025. https://www.saryuindustries.com/blog/which-metal-is-best-conductor-of-electricity

Task 2:

The second part of the project consists in finding the distance that power cables need to be from homes.

Given informations :

- Emax = (80 + A) V/M = 81V/M
- Cables potential from ground = 10 kV

In this problem, we have to assume certain values that are not given. The height of the poles will be assumed to be 10 meters, this value is taken from the average height of poles that are conducting less than 69kV [3].

Step 1:

Using the method of image, we found that:

$$V_0 = \frac{\lambda}{2\pi\varepsilon_0} ln(\frac{2h}{a})$$

$$\lambda = \frac{2\pi\varepsilon_0 V_0}{ln(\frac{2h}{a})} = \frac{2\pi(8.854x10^{-12})(10000)}{8.366} = 6.65\ x\ 10^{-8}\ C/m$$

Step 2:

The Electric field for this case is:

$$E(x) = \frac{\lambda h}{2\pi\varepsilon_0(d^2+h^2)}$$

$$81\ V/M = \frac{\lambda h}{2\pi\varepsilon_0(d^2+h^2)}$$

$$d^2 + h^2 = \frac{\lambda h}{2\pi\varepsilon_0 81}$$

$$d^2 + 10^2 = \frac{2.39x10^4}{81}$$

$$d = \sqrt{\frac{2.39x10^4}{81} - 100} = 14.0m$$

The minimum distance from a house needs to be 14m.

Report:

Before doing this part of the project, I wasn't aware of the securities requirement for installing power lines in residential areas. I believe that this project was my first piece of work in university that actually relates to a real life application.

For this part of the project, I had to search why the exposure to electric fields can lead to health hazards. Like for task 1, I searched online how power lines work and I learned about how to prevent the exposure to electric fields by doing the above calculations.

References:

[3]"Power poles - Great River Energy," *Great River Energy*, 2022.
https://greatriverenergy.com/transmission-and-delivery/power-line-project-faqs/power-poles/